Haoming Shi

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Personal Webpage

Education

Rice University **Houston, TX**
Ph.D. Statistics GPA: 4.0/4.0 Aug 2022 – Present
Ken Kennedy - HPE Cray Graduate Fellowship 2025 - 2026
Advisor: Eric C. Chi and Hengrui Luo

University of California – Santa Barbara **Santa Barbara, CA**
B.S. Financial Math and Statistics GPA: 4.0/4.0 Sep 2018 – June 2022
Dean's Honor Student every quarter.

Research Experience

Rice University **Houston, TX**
Department of Statistics Sep 2023 - Present
Graduate Research Assistant
Current Research Projects:
- Muon-based Fast MCMC Sampler
- Self-Distillation for In-Context Reasoning
- Input Convex Neural Network for Convex Regression
- Asymptotic optimism under tensor regression
- Robust Estimation of Spatiotemporal Models

Lawrence Berkeley National Laboratory **Berkeley, CA**
Applied Mathematics & Computational Research Division May 2025 – Aug 2025
Graduate Research Intern
Supervisor: Xiaoye S. Li
Projects:
- Cheap Gaussian Process surrogate model for efficient cosmology parameter inference

University of Texas MD Anderson Cancer Center **Houston, TX**
Department of Bioinformatics and Computational Biology Aug 2022 – Aug 2023
Graduate Research Intern
Supervisor: Wenyi Wang
Projects:
- Survival-based Clustering of Predictors (SCP) framework
- Bio-imaging analysis on Spatial Tumor Distribution

Stanford Medicine **Stanford, CA**
Center for Asian Health Research and Education June 2021 – June 2022
Undergraduate Research Assistant
Supervisor: Linda Geng and Fumiaki Ikeno
Projects:
- Leading Causes of Death for Japanese Americans
- Misdiagnosis of Rare Diseases among Asian Americans

Teaching Experience

Rice University **Houston, TX**

Teaching Assistant

- STAT 533 (Foundations of Statistical Inference II) Spring 2026
- STAT 613 (Statistical Machine Learning) Spring 2025
- STAT 607 (Statistical Computing) Fall 2024
- STAT 413 (Intro to Statistics Machine Learning) Fall 2023
- STAT 310 (Probability and Statistics) Spring 2023
- STAT 310 (Probability and Statistics) Fall 2022

Student Instructor

- Lecturing for STAT 533 Spring 2026
- Lecturing for STAT 607 Fall 2024
- PhD Qualifying Exam (Statistics and Inference Session) Summer 2024

Presentation and Talks

Publications and Preprints

1. **H. Shi**, E.C. Chi., H. Luo (2026) Asymptotic Optimism for Tensor Regression Models with Applications to Neural Network Compression. arXiv:2603.26048.
2. **H. Shi**, S. Yu, E.C. Chi. (2025) Robust Spatiotemporal Epidemic Modeling with Integrated Adaptive Outlier Detection. Submitted. arXiv:2507.09380.
3. X. Liu, H. Yan, **H. Shi**, E. Montellier, E.C. Chi, P. Hainaut, W. Wang. (2025) Transfer learning for survival-based clustering of predictors with an application to TP53 mutation annotation. Proceedings of The Inaugural Workshop on Frontiers in Statistical Machine Learning.

Selected Talks

1. A Robust Estimation Framework for Spatiotemporal Epidemic Models. *Joint Statistical Meetings. Aug 2024.*

Poster Presentations

1. Robust Spatiotemporal Epidemic Modeling with Integrated Adaptive Outlier Detection. *AI in Health Conference. Sep 2025.*
2. Robust Spatiotemporal Epidemic Modeling with Integrated Adaptive Outlier Detection. *Texas-Louisiana Section of SIAM 8th Annual Meeting. Sep 2025.*
3. A Robust Estimation Framework for Spatiotemporal Epidemic Models. *Institute for Research in Statistics and its Applications. May 2024.*
4. A Robust Estimation Framework for Spatiotemporal Epidemic Models. *43rd Annual Conference of Texas Statisticians. May 2024.*
5. Annotating TP53 Germline Mutations from Patient Time-to-Cancer Diagnosis via Homogeneity Pursuit in Cox Regression. *2023 Leading Edge of Cancer Research Symposium. Nov. 2023.*

Honors and Awards

STAT PhD Student Travel Award. Rice University. Aug 2024, Aug 2025

IMS FSML 2025 Travel Award. The Institute of Mathematical Statistics (IMS). Aug 2025

Highest Honors at Graduation. University of California – Santa Barbara. June 2022

Professional Memberships

Society for Industrial and Applied Mathematics (SIAM). 2025 - Present

American Statistical Association (ASA). 2023 - Present

Volunteer Service

PhD Mentor, PhD Peer Mentoring Program, Rice University — Fall 2025

- Meet with early-stage PhD student to provide suggestions and guidance.

Volunteer, Prospective PhD Student Visit, Rice University — 2023 - Present

- Helped organize and meet with prospective students during onsite visits.

Presenter, PhD Student Seminar, Rice University — Fall 2023

- Presenter recent work at the PhD student seminar.